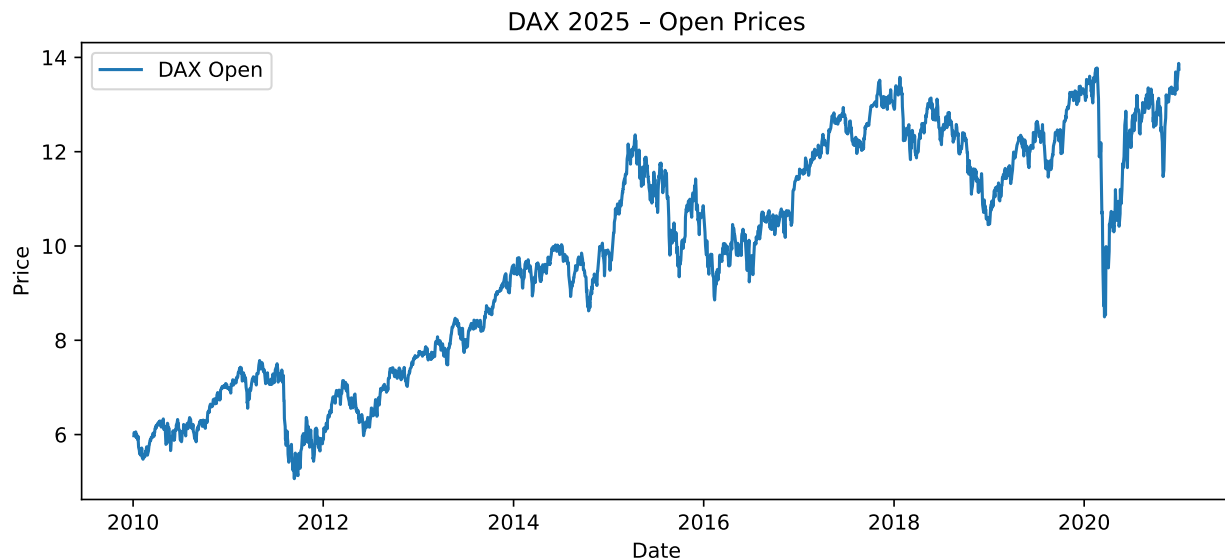


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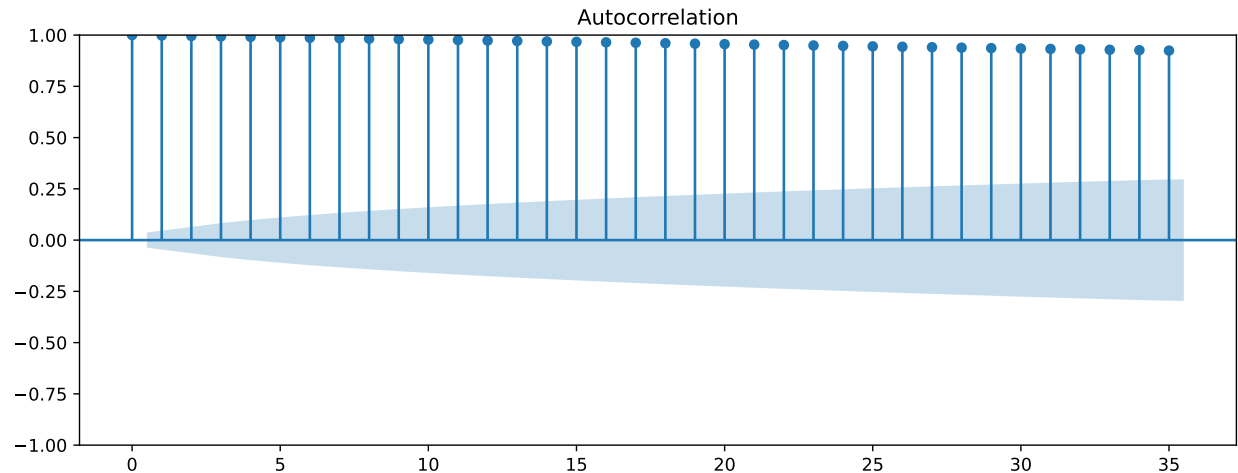
Steven Anthony Leone

Ethan Naegele

We look into the DAX Stock Index in Germany in this project. We look at the daily stock price for the DAX index for every day in 2025 from a time series analysis perspective, and we see how accurate we can make a model that predicts future values based off of its lags.

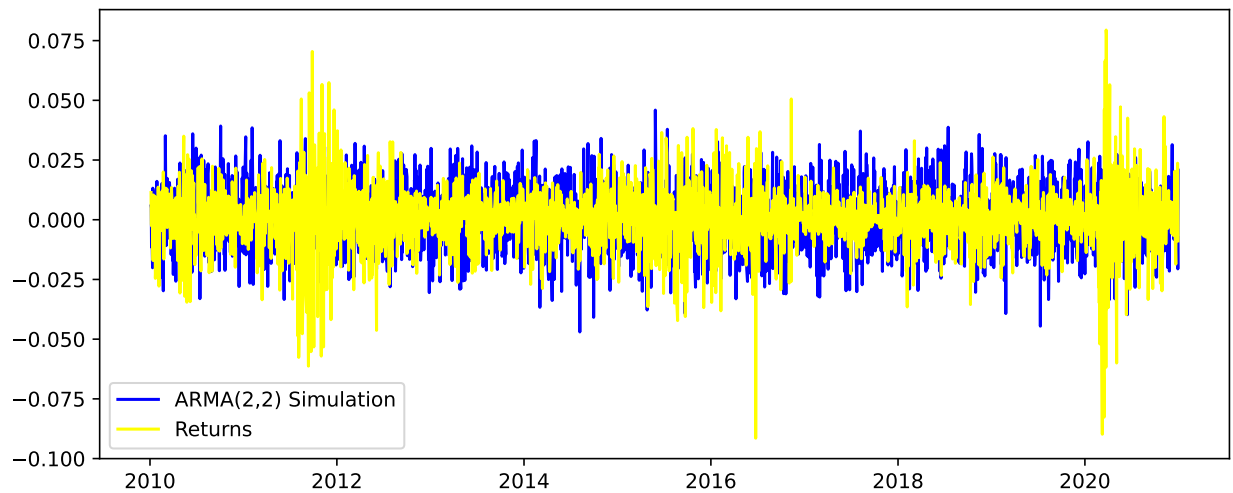


The data above shows roughly nonstationary data. We look at the autocorrelation function below to see how much lags depend on each other. In general, what we can tell is that the first point has a high correlation with the lags that follow.



ARMA

Let's train a model to accurately reflect the data. First, let's try an autoregressive model - we notice that the model has lags that correlate well with the first 10, so, an autoregressive model should model it well. ARMA models do not work well with nonstationary data; we counter this with an ARIMA model that handles differencing.

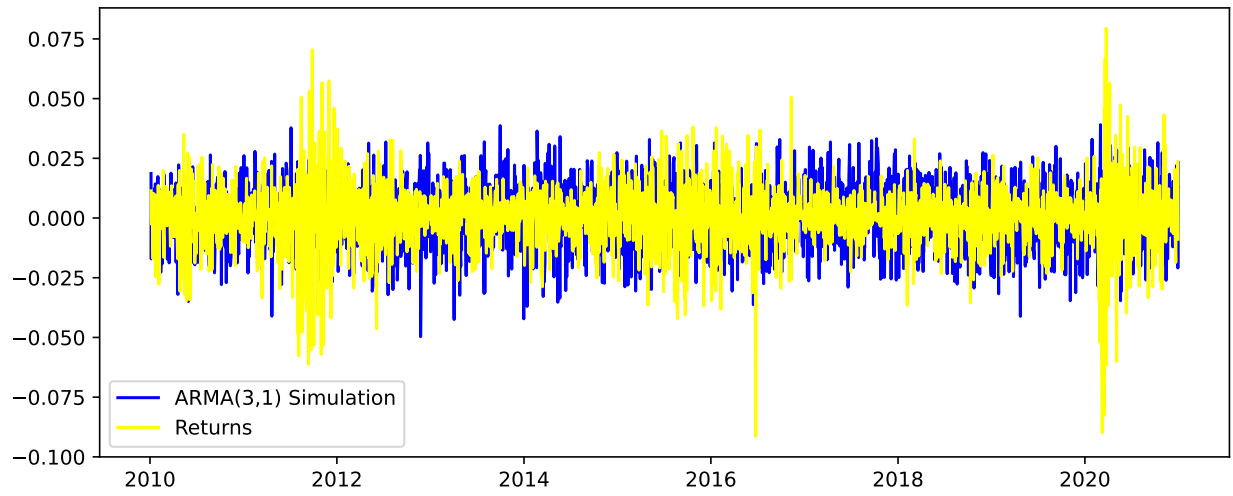


We look at the AIC values of the ARMA models below. The smallest AIC value is -3961.6, for ARMA (3, 1).

We graph an ARMA(3, 1) model below. It is not much different from ARMA(2, 2).

	MA(1)	MA(2)	MA(3)	MA(4)
AR(1)	-3965.5	-3965.7	-3972.9	-3973.5
AR(2)	-3962.0	-3966.6	-3972.8	-3971.7
AR(3)	-3961.6	-3970.7	-3974.7	-3972.2
AR(4)	-3972.8	-3974.8	-3972.4	-3968.0

Table 1: ARMA Model Comparison



SARMA

TODO ...

References